

Surface Mapper

Huxe · 10-Trace Exercise · maps observed failures onto pipeline stages to direct measurement

↶ Re-tag

↓ Export

Update T-01 Brief →

Traces Sampled

10

from 186 release-gate cases · profile-stratified

Tags Emitted

27

across 7 surfaces + Audio-Native

Causal Roots Identified

4

S1 (×2) · S6 · TTS (S7)

Evidence Floor

Review-derived REV for delivery; SIM elsewhere

Trace × Surface heat map — each cell shows tag count and severity on that surface

clean

minor

failure

causal root

Trace	S1 INGESTION	S2 NORMALIZE	S3 INDEX	S4 RETRIEVE	S5 REASON	S6 GENERATE	S7 TTS	S8 GUARD	S9 DELIVER	A AUDIO-NATIVE	Outcome
eval-2847 Profile A · lawyer 14-msg thread + standup	2	1	.	1	.	1	1	.	1	4	FAIL
eval-2851 Profile B · founder slack-export + investor cal	1	.	.	1	1	2	1	.	.	2	FAIL
eval-2856 Profile C · researcher arxiv + 30-msg dm	1	.	.	1	.	1	.	.	.	2	WARN
eval-2860 Profile A · lawyer 5-msg client thread	.	.	.	.	.	.	1	.	.	1	PASS
eval-2864 Profile B · founder board-deck + 12 cal	1	1	.	1	.	1	1	.	.	1	WARN
eval-2869 Profile A · lawyer 22-msg thread + 5 attach	3	1	.	1	.	2	1	.	1	3	FAIL
eval-2873 Profile C · researcher pdf + colleague chat	.	.	.	.	.	1	1	.	.	1	WARN
eval-2878 Profile B · founder invest email + roadshow	1	1	.	1	.	1	2	.	.	2	FAIL
eval-2882 Profile A · lawyer deposition prep · 8 msg	1	.	.	.	.	1	1	.	.	1	WARN
eval-2887 Profile C · researcher grant memo + cal block	.	.	.	.	.	.	1	.	.	.	PASS
TAG COUNT	10 7/10 traces	3 3/10	0	5 5/10	1 1/10	9 8/10	9 9/10	0	2 2/10	17 9/10	5 FAIL 3 WARN 2 PASS ✓

Surface ranking — ordered by tag count; informs measurement priority

A

Audio-Native

17 tags · 9/10 traces

S1

Ingestion

10 tags · 7/10 · 2 roots

S6

Generator

9 tags · 8/10 · 1 root

S7

TTS

9 tags · 9/10 · 1 root

S4

Retriever

5 tags · 5/10

S2

Normalizer

3 tags · 3/10

S9

Delivery

2 tags · 2/10 REV

C1

Ingestion Fidelity · root for HC-01 cascade

37%

C2

Faithfulness · downstream of S1 collapse

26%

C5

Output Quality · Audio-Native dominates

22%

C3

Factuality / consistency · secondary

15%

Proposed T-01 Brief deltas — surfaces this exercise demands we measure

+ ADD Audio-Native dimension panel (6 dims, listen-aligned)

17 tags across 9/10 traces · current Brief measures transcript only · Owner: Eval Lead · ETA: this sprint

Ingestion: "email_thread_truncation" tag {single-tag}

Ingestion: split into `thread\_collapse`, `attendee\_drop`, `chunk\_boundary` — each with its own canary set

3 of 10 traces show S1 as HC-01 causal root. Single-tag treatment under-counts cause vs. symptom.

TTS: SIM evidence-floor

TTS: REV evidence floor for entity pronunciation (top-200 named entities)

Mispronunciations confirmed via human listen; SIM judge missed them entirely.

+ ADD Delivery monitor: REV CDN/playback drop sampler (1% live traffic)

2/10 traces had silent audio drops invisible to transcript. Cannot be SIM'd - needs LIVE path.

Retriever scoring: pass/fail-boolean

Retriever scoring: recall@k against profile-graph ground truth

5/10 traces show retriever miss; current boolean hides ranking degradation.

Reject all

Edit before sending

Send 5 deltas → T-01 Brief →